MY ACADEMIC RESUME

Sidi Mohammed Kaddour

R&D Engineer, Inria (STACK team)

medkaddour.github.io | Nantes , Pays de la Loire , FR

Research and development engineer working on the edge-to-cloud continuum: application-level observability, feedback loops, and coordinated deployment of distributed systems. PhD in networks and distributed systems, with a background in machine learning for smart energy and in industrial ERP engineering.

WORK

2024.12 - 2026.11

Nantes, France

R&D Engineer

Inria, STACK team

CoAnsible: extending Ansible with the coordination model of Concerto for adaptive distributed deployments.

- Translation of Ansible playbooks into Concerto assemblies, with automated dependency extraction
- Grid'5000 provisioning tooling driven from Ansible inventories (EnOSlib)
- Evaluation of coordinated reconfiguration on multi-node deployments

2023.12 - 2024.11

Nantes, France

R&D Engineer

IMT Atlantique, STACK team

Feedback mechanisms and application-level observability for Edge-to-Cloud applications.

- Distributed video processing pipeline: motion detection at the edge, YOLO recognition in the cloud
- Instrumentation with OpenTelemetry, Prometheus, Grafana and distributed tracing
- SLO-aware feedback loop with fault injection via Chaos Mesh, deployed on k3s over Grid'5000
- Published at UCC '25 (ACM/IEEE)

2019.06 - 2019.08

Bournemouth, UK

Visiting PhD Student

Bournemouth University

Smart energy solutions using machine learning, hosted by Prof. Abdelhamid Bouchachia.

2018.07 - 2023.09

Tlemcen, Algeria

ERP Project Manager & Senior Odoo Developer

Groupe Kherbouche (InterEntreprise)

Led the ERP system of the group, from module development to production operations and team coordination.

- Custom OpenERP/Odoo modules for accounting, inventory, HR and sales
- Linux server and PostgreSQL administration, backups and upgrades
- Requirements gathering with department heads, release and change management

2017.09 - 2020.09

Tlemcen, Algeria

Temporary University Teacher

University of Tlemcen

Taught Advanced Operating Systems, Cloud Computing, and Signal Processing at Bachelor and Master level.

2014.04 - 2016.11

Tlemcen, Algeria

Delphi Developer (remote)

SoftBuilder

ERBuilder data modeler: database reverse engineering and SQL generation across relational engines.

EDUCATION

2016.11 - 2023.11

Tlemcen, Algeria

PhD

University of Tlemcen

Networks and Distributed Systems - thesis: Soft Sensors for Complex Systems

2014.09 - 2016.06

Tlemcen, Algeria

Master

University of Tlemcen

Networks and Distributed Systems

2011.09 - 2014.06

Tlemcen, Algeria

Bachelor

University of Tlemcen

Computer Science

2008.09 - 2011.07

Nedroma, Algeria

Baccalaureate

Mostafa Mostfai High School

Sciences

PUBLICATIONS

2025.12.01

Application level observability for adaptive Edge to Cloud continuum systems

UCC '25 - 18th IEEE/ACM International Conference on Utility and Cloud Computing

SLO-aware feedback loops built on OpenTelemetry, Prometheus, k3s and Chaos Mesh. Preprint: arXiv:2601.14923

2025.06.16

Multi-provider capabilities in EnOSlib: driving distributed system experiments on the edge-to-cloud continuum

DAIS 2025 - 25th IFIP WG 6.1 International Conference (DisCoTec)

Synchronized, reproducible experiments spanning several testbeds. Open access: hal-05052776

2024.06.03

Proteus: Towards Intent-driven Automated Resource Management Framework for Edge Sensor Nodes

FlexScience '24 workshop, ACM HPDC

Intent-based resource management for edge sensor nodes. Open access: hal-04775138

2024.02.27

Event Detection for Non-intrusive Load Monitoring using Tukey's Fences

arXiv preprint

FFT-based event detection with statistical fences for appliance-level energy monitoring.

2023.10.01

Soft Sensors for Complex Systems

PhD thesis, University of Tlemcen / STIC Laboratory

2021.07.01

Electricity Consumption Data Analysis Using Various Outlier Detection Methods

International Journal of Software Science and Computational Intelligence (IJSSCI)

Isolation Forest, One-Class SVM and K-Means compared on real consumption traces.

PROJECTS

2024.01 - Present

Edge-to-cloud video processing

Distributed surveillance pipeline with edge motion detection, cloud YOLO recognition, full instrumentation and an SLO-driven feedback loop. Artifact of the UCC '25 paper.

- OpenTelemetry + Prometheus instrumentation
- k3s deployment on Grid'5000
- Fault injection with Chaos Mesh

2025.01 - Present

CoAnsible

Bringing Concerto's coordination model into Ansible: playbook dependency extraction, assembly generation, and Grid'5000 provisioning tooling.

2024.06 - 2025.06

EnOSlib multi-provider

Contributor to the multi-provider extension of EnOSlib, enabling synchronized experiments across several testbeds (DAIS 2025).

2025.01 - 2025.12

Conference websites (UCC 2025, BDCAT 2025)

Hugo-based sites for two IEEE/ACM conferences held in Nantes, deployed through GitLab CI/CD.

2024.09 - Present

Teaching material: DevOps and Middleware

Hands-on labs on Ansible, Docker, Kubernetes and OpenStack for Master students and continuing education.

VOLUNTEER

2025.01 - 2025.12

Co-Web Chair

IEEE/ACM BDCAT 2025

12th IEEE/ACM International Conference on Big Data Computing, Applications and Technologies, Nantes, France.

2025.01 - 2025.12

Co-Web Chair

IEEE/ACM UCC 2025

18th IEEE/ACM International Conference on Utility and Cloud Computing, Nantes, France.

- Built and maintained the conference website (Hugo, GitLab CI/CD)
- Published the technical program and schedule

SKILLS

ORCHESTRATION & INFRASTRUCTURE

Ansible
Concerto
Kubernetes / k3s
Docker
Helm
OpenStack
CI/CD

OBSERVABILITY

OpenTelemetry
Prometheus
Grafana
Zipkin
Jaeger
Chaos Mesh

EXPERIMENTATION

Grid'5000
EnOSlib
Reproducible experiments
Performance evaluation

PROGRAMMING

Python
C/C++
Java
Delphi/Pascal
Bash
SQL
JavaScript

MACHINE LEARNING

scikit-learn
Keras/TensorFlow
OpenCV
Time series
Anomaly detection

CERTIFICATES

ISO 9001 TRAINING

InterEntreprise 2022-01-01

ITIL V4 FOUNDATION

AXELOS 2021-01-01

ISO 19011 INTERNAL AUDITOR CERTIFICATION

InterEntreprise 2021-01-01

LANGUAGES

ARABIC

Fluent

ENGLISH

Proficient

FRENCH

Intermediate

INTERESTS

RESEARCH

Edge-to-cloud continuum
Adaptive systems
Observability
Reproducibility

APPLIED ML

Smart energy
NILM
Anomaly detection

PERSONAL

Drawing
Mobile game development
Travel